The Eclipse Rich Ajax Platform

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The Innoopract pitch

Integration & delivery

Yoxos provides
a plugin library
and tools that enable
enterprises to manage
and deliver
eclipse technology

Technology

Open Source
Project that extends
the reach of
Eclipse components
to the web

Services

Helping corporations to implement Eclipse strategies,
offering consulting, development, support and training
We offer

RAP consulting, developer support, production support

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Agenda

What is RAP

How does it work

Delveloper‘s view

Single Sourcing RAP/RCP

Demo

Outlook

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What we like about Eclipse

plug-ins, plug-ins, plug-ins – bundles too ...

OSGi specifies a dynamic component model:

- Module – encapsulation and declaration of dependencies
- Life Cycle – API for life cycle management
- Service Registry – providing functionality to other bundles
- Security layer - limit bundle functionality to pre-defined capabilities

The Eclipse OSGi implementation is provided by the Equinox Project

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What we like about Eclipse (cont'd)

contribution to a common ui concept

- Standard Widget Toolkit (SWT)
  - delivers native widget functionality for the Eclipse platform in an operating system independent manner
- JFace
  - sits on top of SWT and provides classes for handling common UI programming tasks
- Workbench
  - is responsible for the presentation and coordination of the user interface

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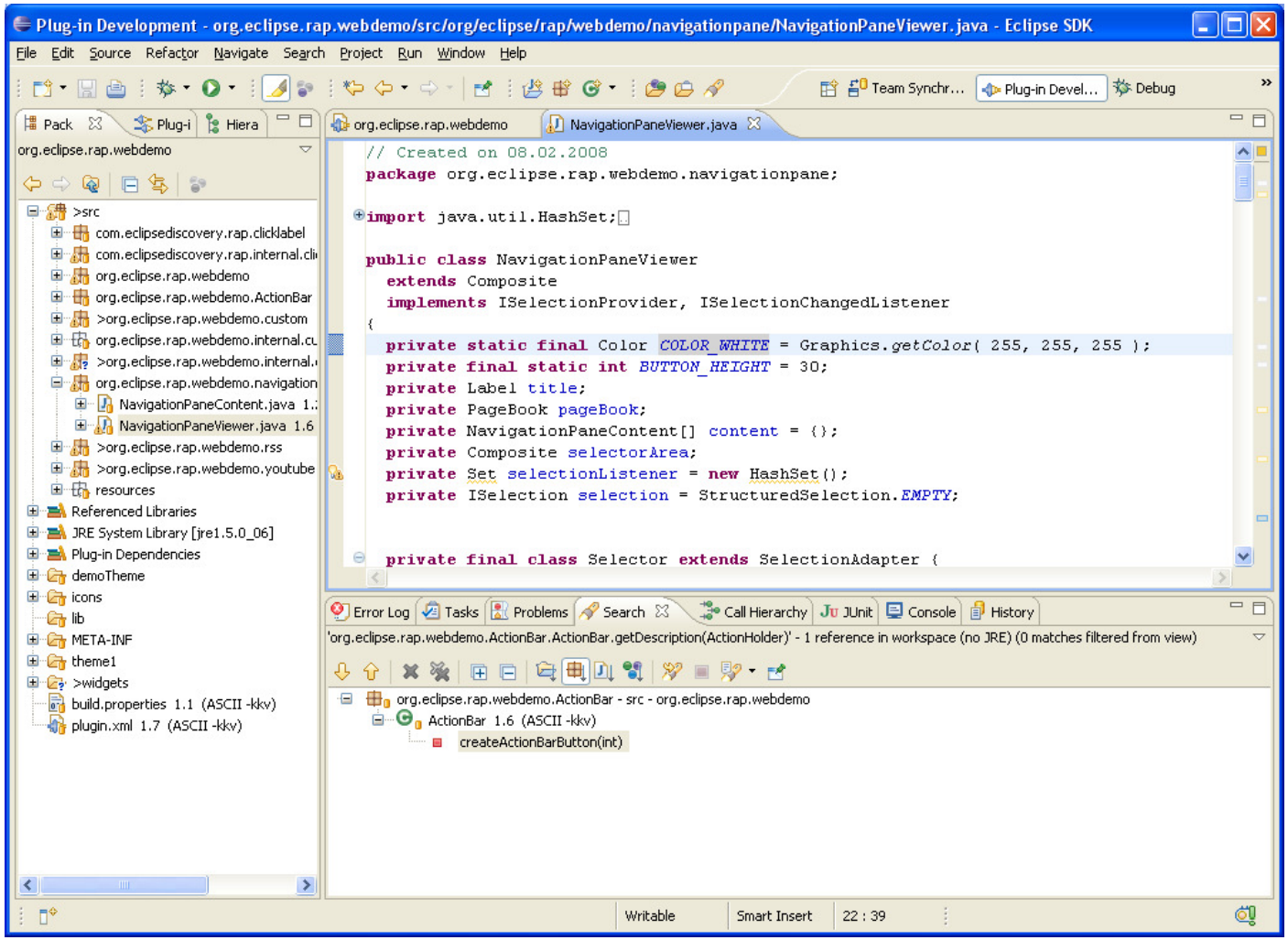
What we like about Eclipse (cont'd)



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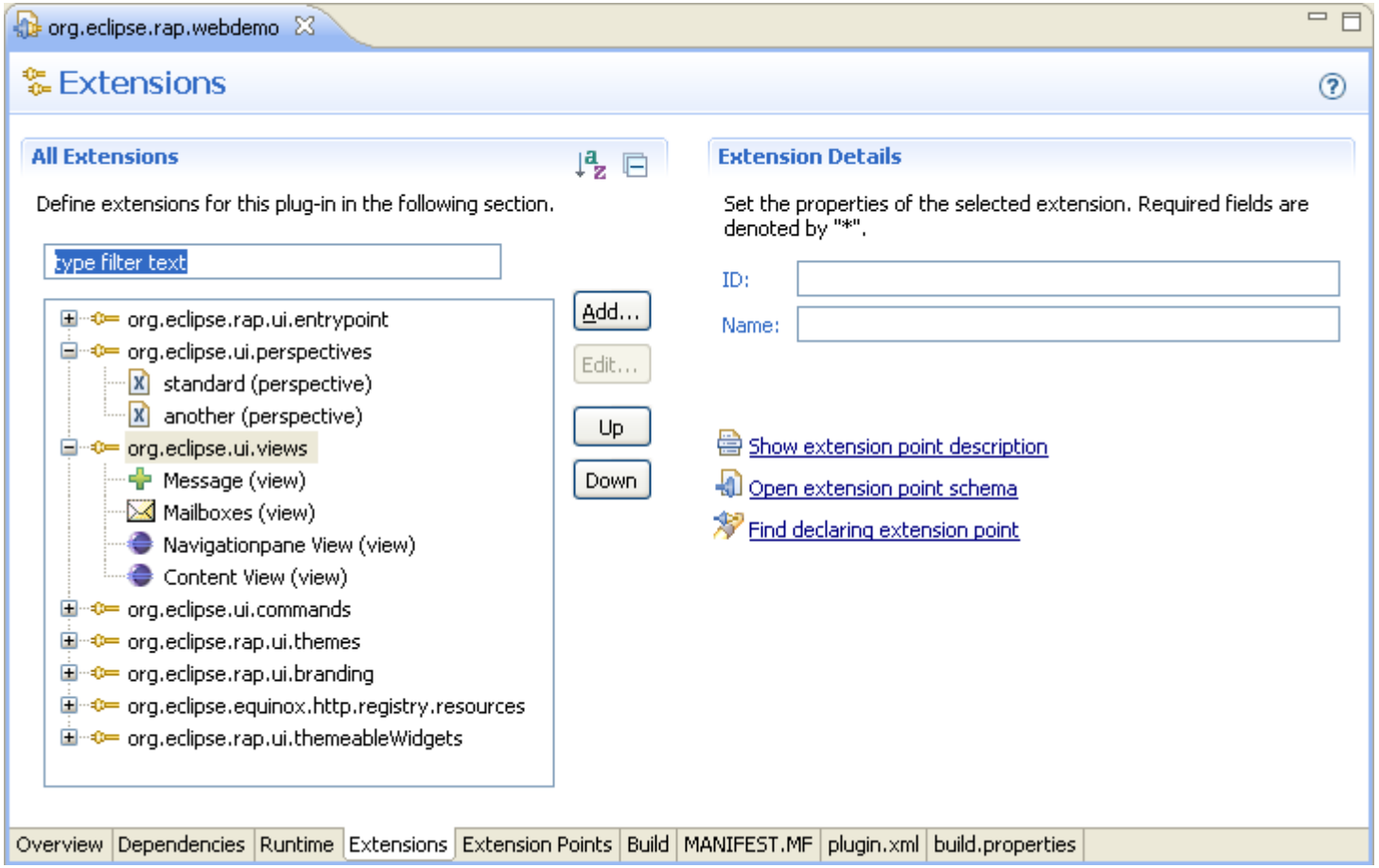
What we like about Eclipse (cont'd)



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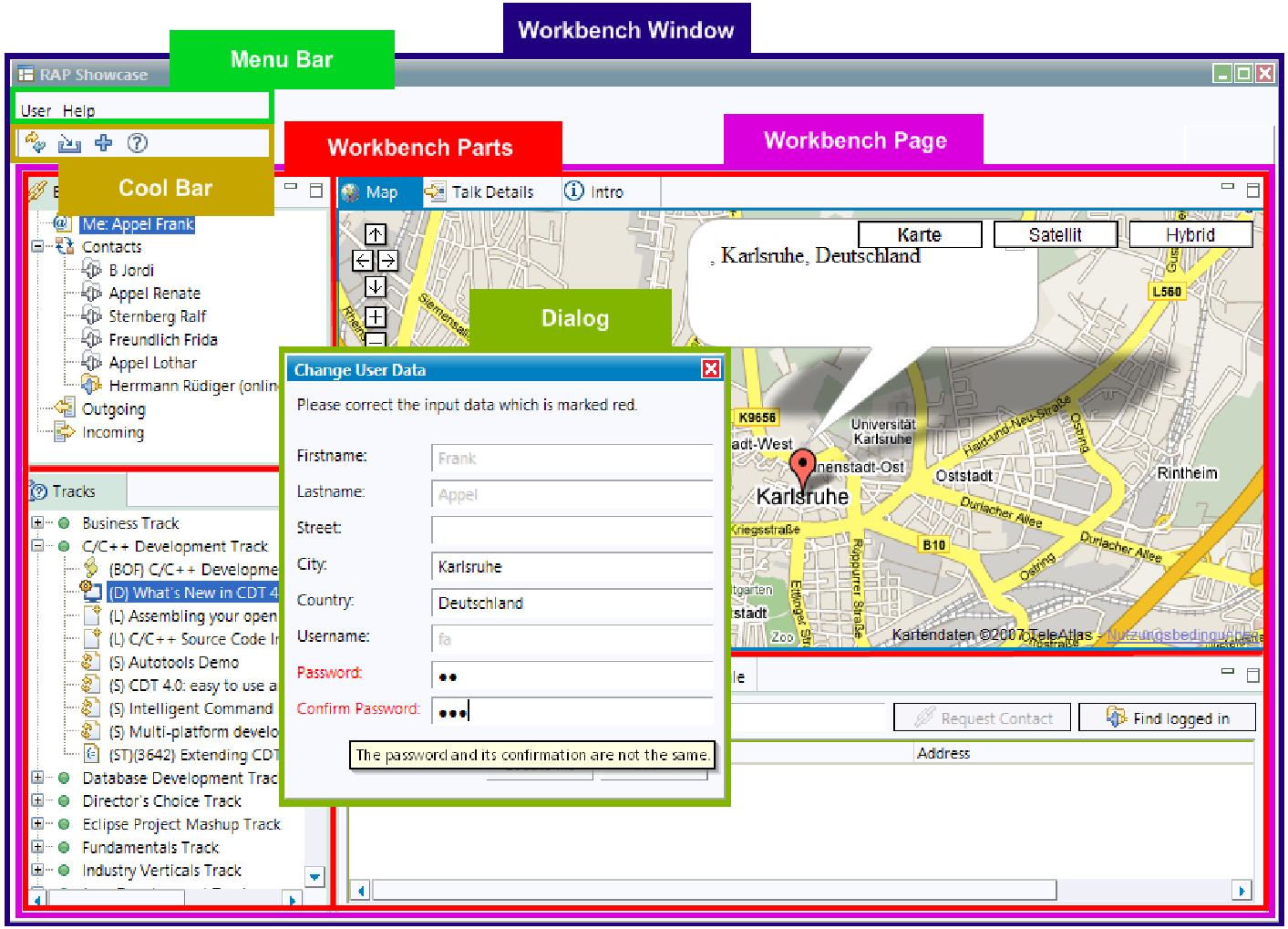
How does that translate to the web?



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How does that translate to the web? (cont'd)

RAP - enabling plug-in reuse

- 70% - 90% reuse is possible
- RAP provides only a subset of RCP!
- needs separation of code that is not compatible
  can be addressed with fragments
- application needs to become multi-user enabled

getting all the things we like about Eclipse

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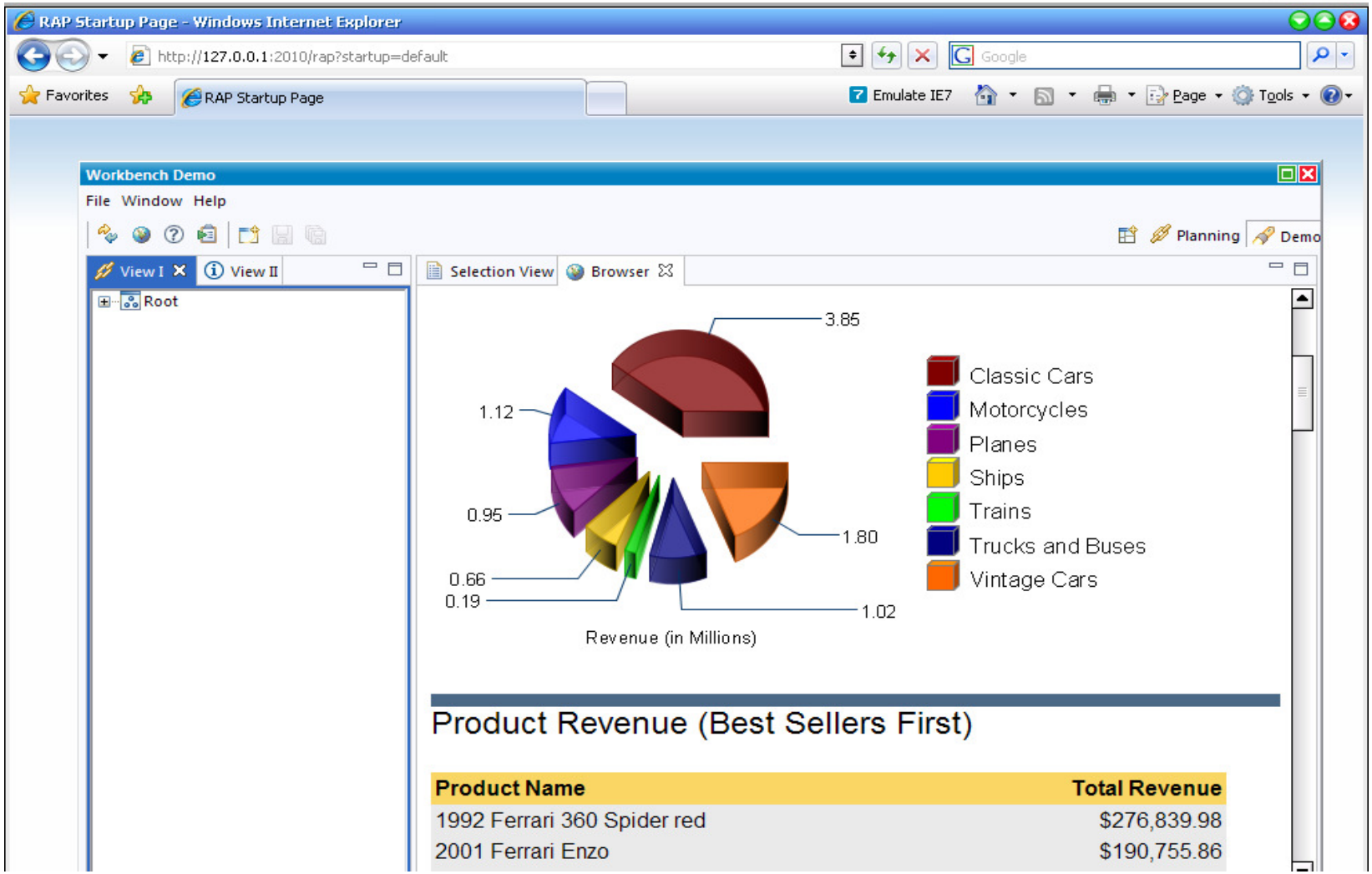
Nice idea – but I don‘t like RCP in a browser



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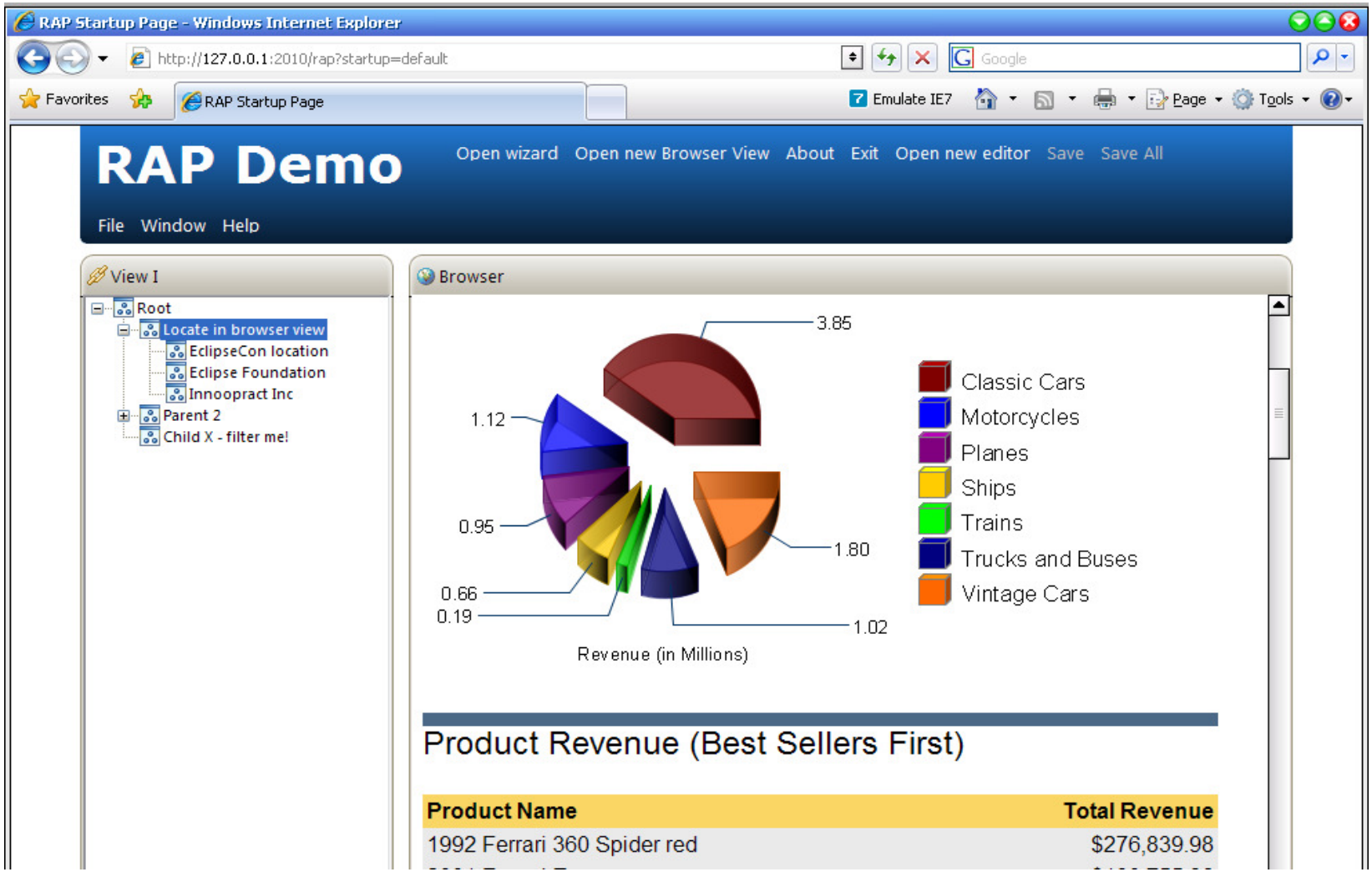
O.K. - but it does not have to be this way



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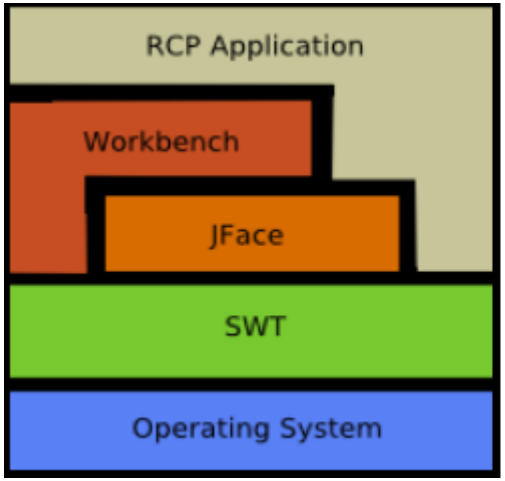
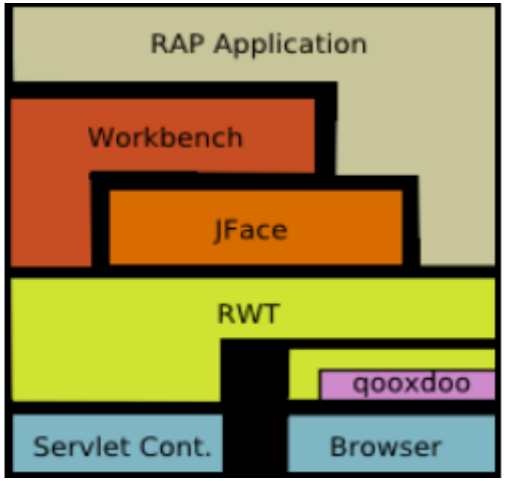
How does it work?





- replacing SWT with an implementation that can render to browsers
- everything else is pretty much the same
- RWT uses qooxdoo Javascript library to render widgets in the browser
- differences:
  - multi-user environment: OSGi bundles shared between sessions
  - Additional API for web specifics

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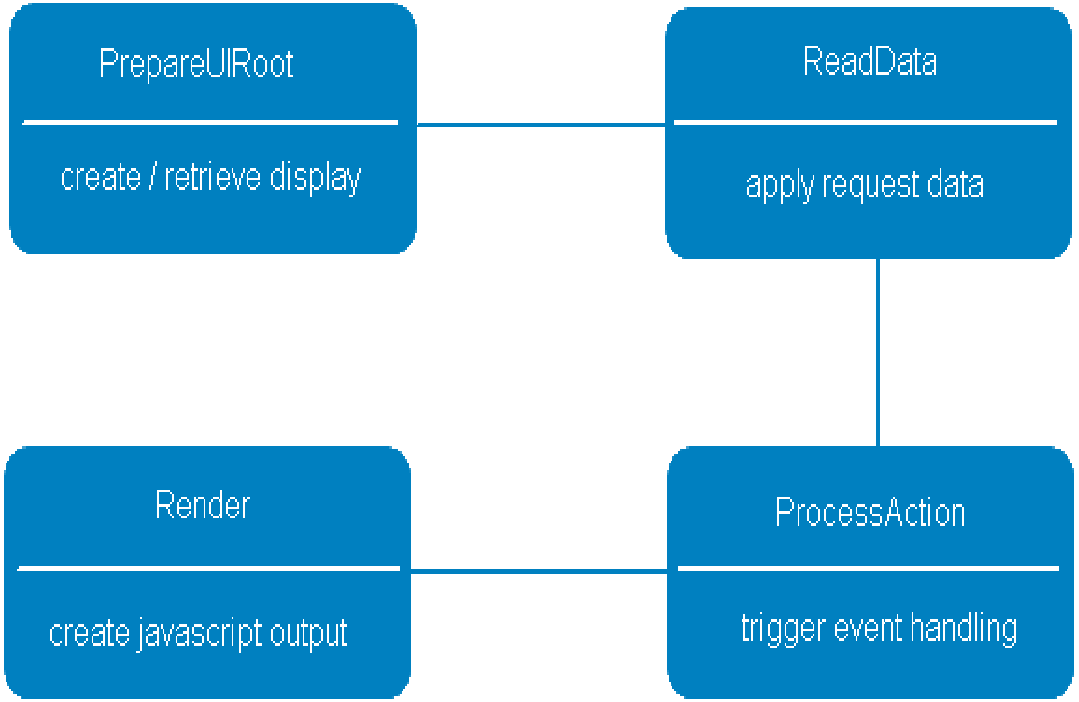
How does it work? (cont'd)

The Request Lifecycle



- At the end of the ***ReadData*** phase all widget attributes are in sync with the client
- These attributes are preserved for later comparison
- During the ***ProcessAction*** *phase* attribute changes may occur
- The ***Render*** Phase compares the widget attributes with their preserved values and submits only the delta to the client

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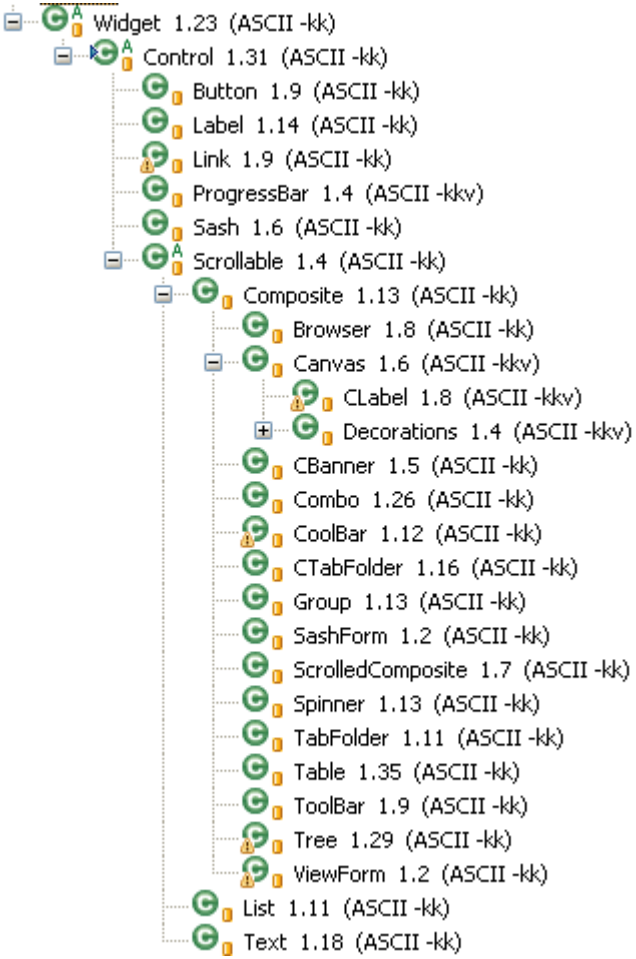
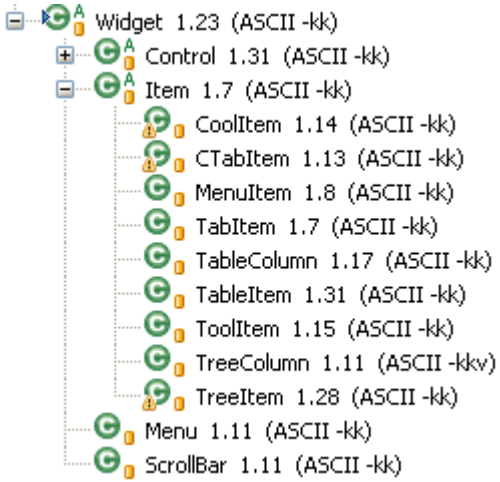
RWT Widgets





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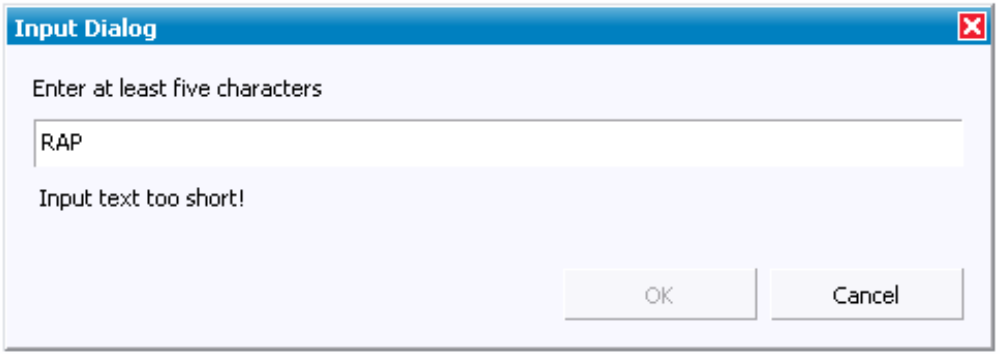
RWT Events and Listeners

- available Listener types:
  - SelectionListener
  - ControlListener
  - ShellListener (supports doit flag now)
  - MenuListener
  - ModifyListener
  - VerifyListener
  - MouseListener
- more in future:
  - KeyListener
  - Drag/DropListener



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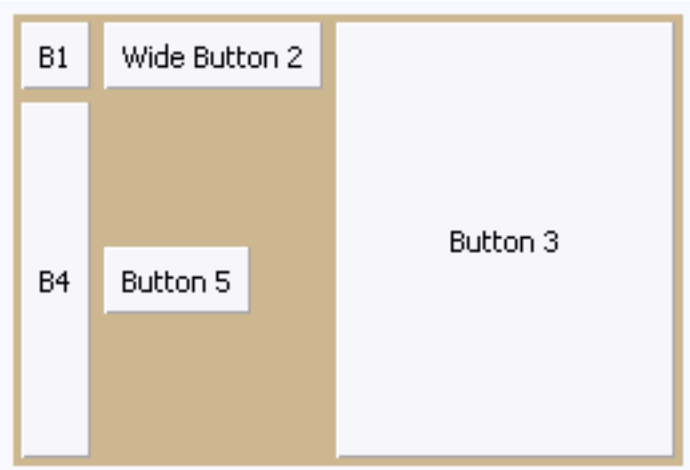
RWT Layouts

- all usual layouts:
  - GridLayout,
  - RowLayout
  - FillLayout
  - FormLayout
  - StackLayout
  - and a lot more ...



- mostly verbatim copies of SWT (OS independent)
- layout algorithms work exactly as in SWT
- layouts are computed on the server, e.g. after a Shell has been resized

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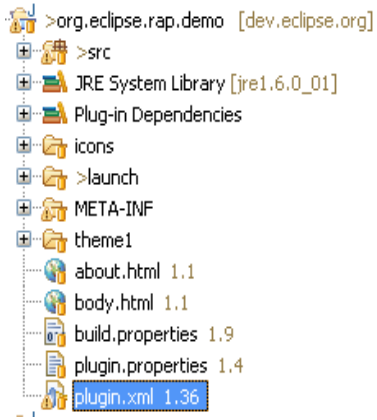
Developer's view on a RAP application

- Startup involves extensions or implementations of the following types:
  - WorkbenchAdvisor
  - WorkbenchWindowAdvisor
  - ActionBarAdvisor
  - IPerspectiveFactory
- These types are quite familiar for RCP developers and serve the same purpose as their RCP equivalents




```
<extension
    id="org.eclipse.rap.demo.demoentrypoint"
    point="org.eclipse.rap.ui.entrypoint">
  <entrypoint
        class="org.eclipse.rap.demo.DemoWorkbench"
        id="org.eclipse.rap.demo.entrypoint1"
        parameter="default"/>
</extension>
```

```
public class DemoWorkbench implements IEntryPoint {

  public int createUI() {
    Display display = PlatformUI.createDisplay();
    return PlatformUI.createAndRunWorkbench( display, new DemoWorkbenchAdvisor() );
  }
}
```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Developer's view on a RAP application (cont'd)

- JDT: content assist, refactoring, etc., PDE, Javadoc available
- developer does not get in touch with Javascript, CSS, HTTP ...

```
org.eclipse.rap.demo        *DemoTreeViewPart.java
 * Copyright (c) 2002-2006 Innoopract Informationssysteme GmbH.

package org.eclipse.rap.demo;

import java.util.ArrayList;

public class DemoTreeViewPart extends ViewPart implements IDoubleClickListener {

  private TreeViewer viewer;

  public void createPartControl( final Composite parent ) {
    viewer = new TreeViewer( parent );
    viewer.setLabelProvider(new DecoratingLabelProvider(new LabelProvider(),
                                                        new LeafStarLabelDecorator()));
    viewer.setc
    viewer.
    viewer.
    getSite
  }

  private f
    public
      if(

setCellModifier(ICellModifier modifier) void - ColumnViewer
setChildCount(Object elementOrTreePath, int count) void - T
setColumnProperties(String[] columnProperties) void - Colum
setComparator(ViewerComparator comparator) void - Structu
setComparer(IElementComparer comparer) void - Structured
setContentProvider(IContentProvider provider) void - TreeVi

Press 'Ctrl+Space' to show Template Proposals

The AbstractTreeViewer implementation of this method checks to ensure that the content provider is an ITreeContentProvider.
```


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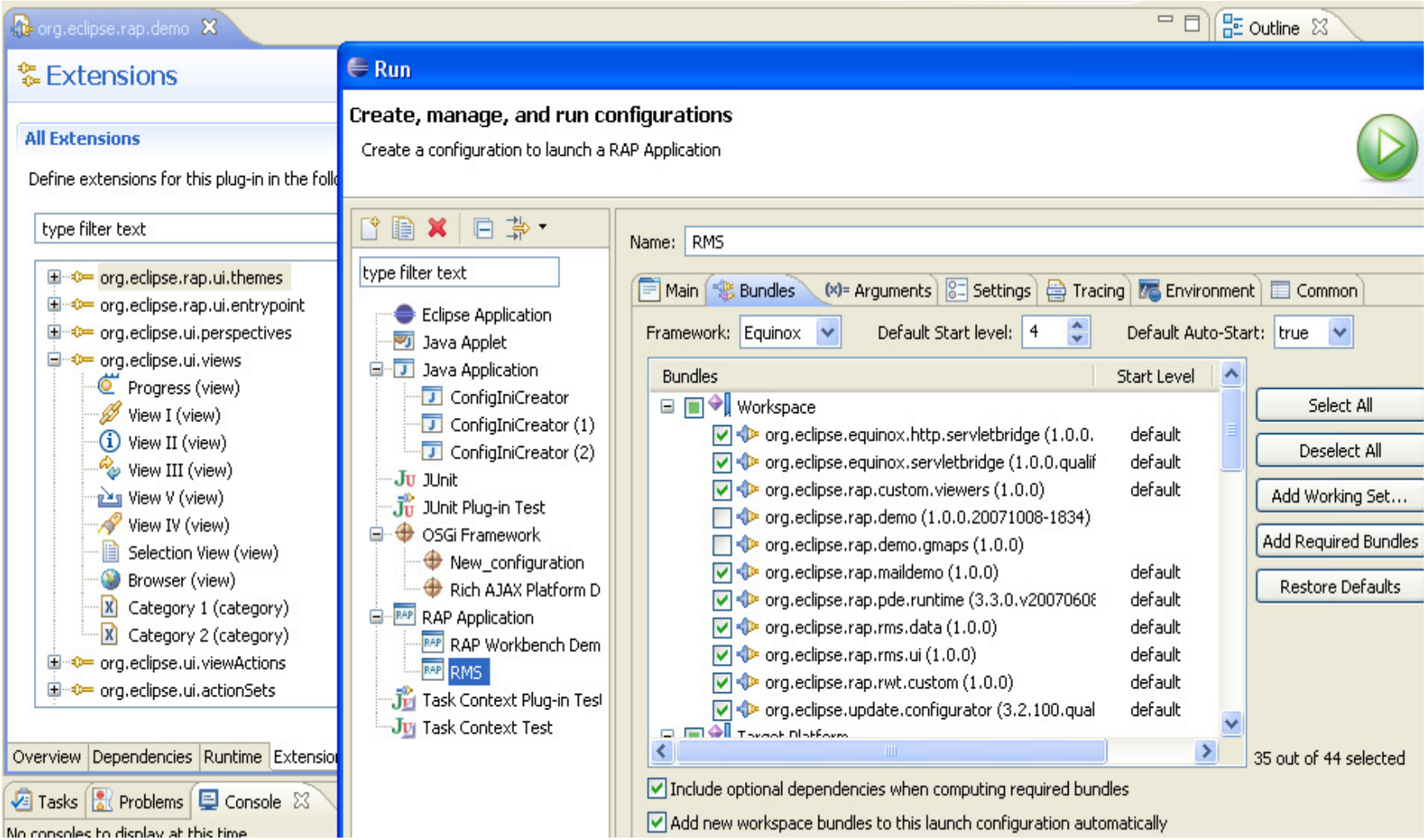
Developer's view on a RAP application (cont'd)



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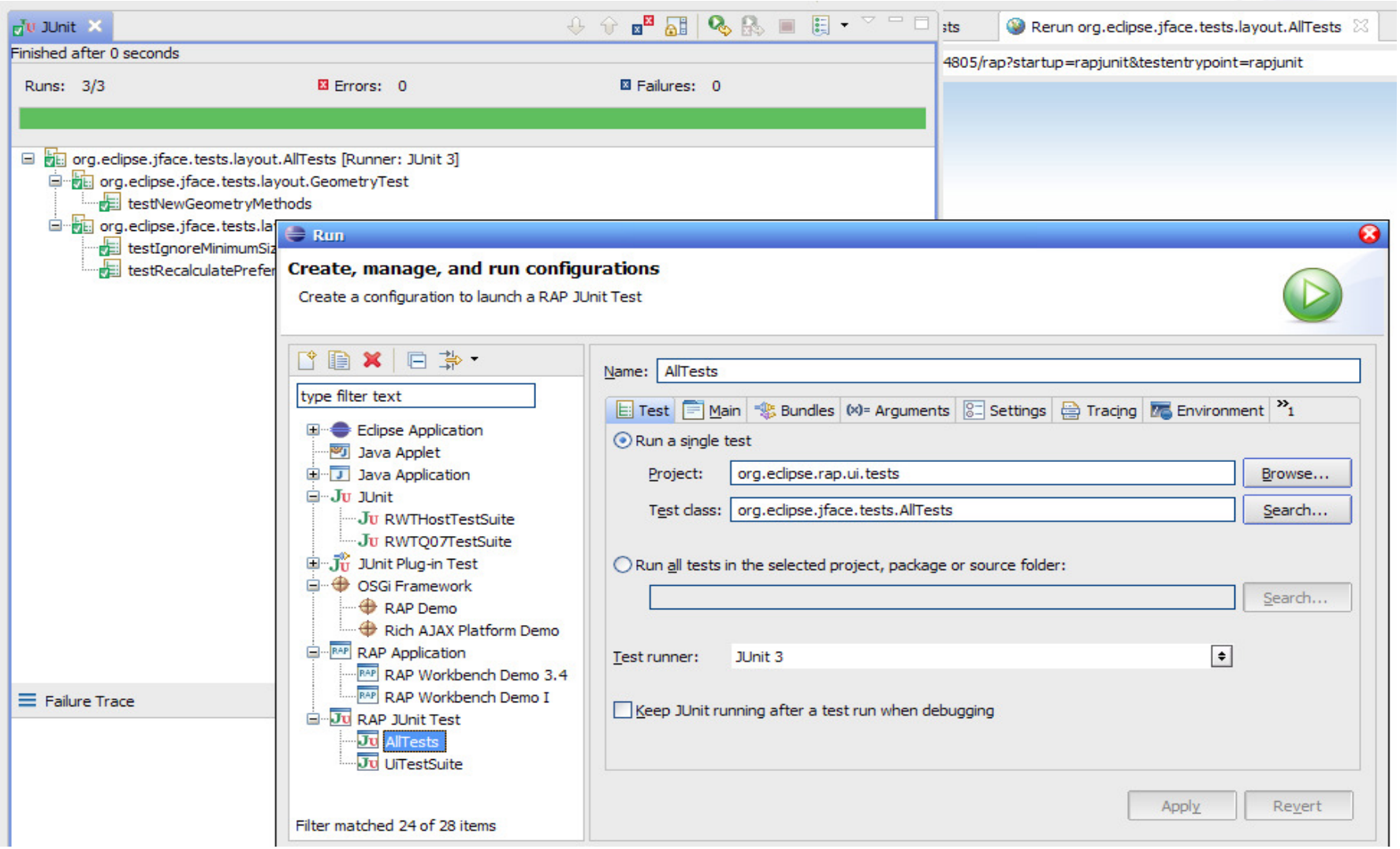
Developer's view on a RAP application (cont'd)



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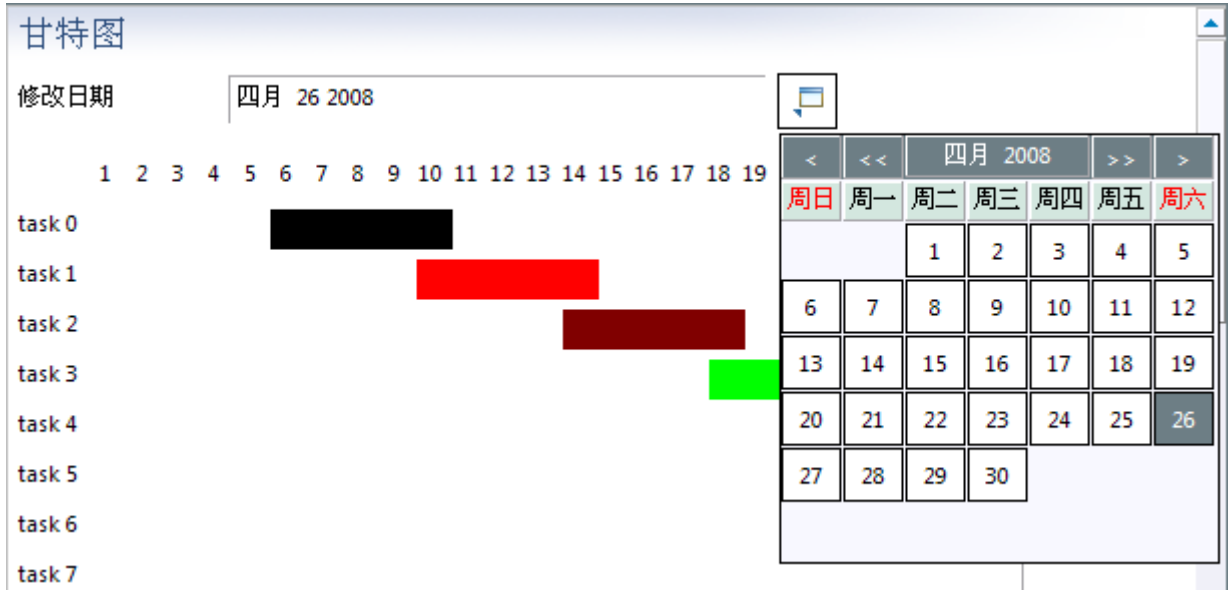
Composed Widgets – just like SWT



- composition of existing widgets
- custom layouts, event handling
- application developers uses composition as a control

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Custom Widgets – the shortcut

Use the browser widget!

```
public YoutubeShell( final Display display ) {
  this.display = display;
  bgColor = display.getSystemColor( SWT.COLOR_BLACK );
  ...
  createShield();
  createShell();
}

public void setId( final String id ) {
  browser.setText( getHtml( id ) );
}

private static String getHtml( final String id ) {
  String html = "<html><body>"
    + "<object width=\"425\" height=\"373\">"
    + "<param name=\"movie\" value=\"http://www.youtube.com/v/" + id
    + "&rel=1&border=1\"></param>"
    + "<param name=\"wmode\" value=\"transparent\"></param>"
    + "<embed src=\"http://www.youtube.com/v/" + id + "&rel=1&border=1\"
```


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Custom Widgets – the shortcut

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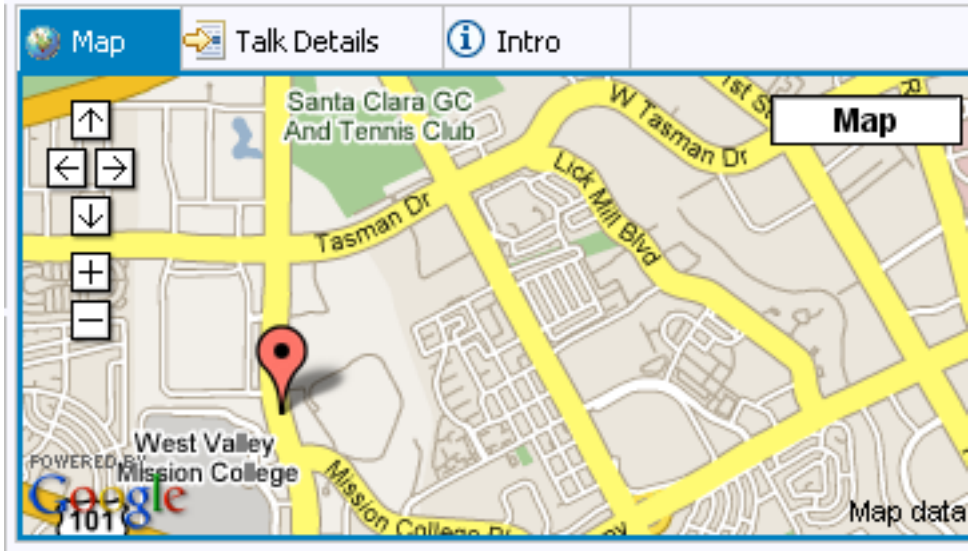
Custom Widgets - the real thing

- like in SWT, requires good knowledge of the platform
- component developer needs javascript, qooxdoo and RAP knowledge
- allows do embed all sort of client side technologies: JS frameworks, Flash, Applets, ...
- application developer simply uses java API
- tutorial in RAP Help




```
GMap map = new GMap( shell, SWT.NONE );
map.setAddress( "5001 Great America Pkwy, Santa Clara" );
```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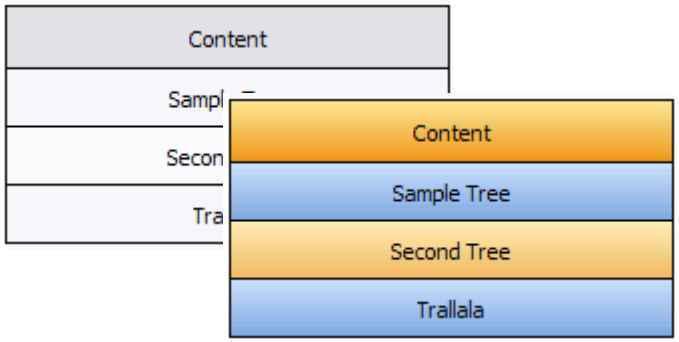
Web Look & Feel: Theming

- objective: allow for a custom look of web applications
- predefined properties of widgets can be customized
- Dimensions, Colors, Borders, Fonts, Images
- css syntax
- themeable custom widgets

```
* {
  backgound-color: white;
}

Button[PUSH], Button[TOGGLE] {
  border: 2px solid blue;
  color: rgb( 17, 23, 103 );
  background-color: #f9f9f9;
}

Button[PUSH]:hover, Button[TOGGLE]:hover {
  background-color: white;
}
```



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Web Look & Feel: Workbench Technology

- PresentationFactory
- WorkbenchWindowAdvisor
- ContributionItems

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Single Sourcing RCP/RAP

Differences between RAP and RCP

- RAP applications are client-server applications with a browser as the client (Thin Client) and the OSGi infrastructure on the server
- OSGi-Bundles are shared between sessions (in general there are 3 scopes on the server: application-, session- and request-scope)
- Restrictions of the distributed environment do not allow to provide the complete API of SWT
  - no GC
  - no constructors, dispose for Font, Image and Color

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Single Sourcing RCP/RAP (cont'd)

More differences – multi users

SINGLETONS are shared between ALL users in RAP

- RAP provides a class called SessionSingletonBase, that can be subclassed to provide Singletons by Session:

```
public class MySessionSingleton extends SessionSingletonBase {
  public static MySessionSingleton getInstance() {
    return(MySessionSingleton)getInstance(MySessionSingleton.class);
   }
 }
```

- access to SessionSingletons is simple within the UI thread, but need special care in background processes (jobs)

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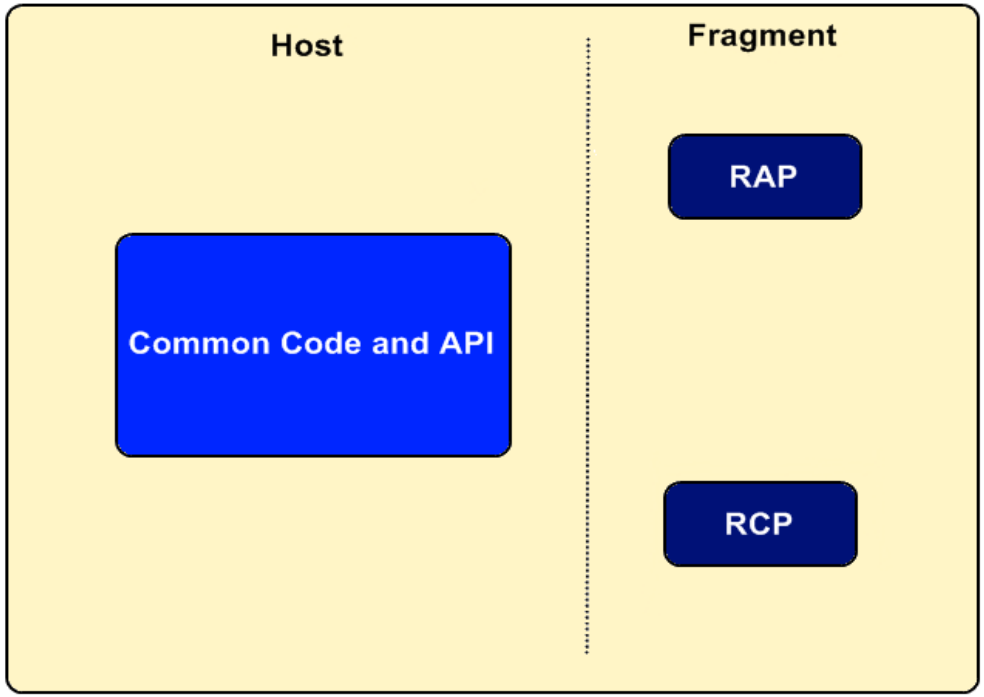
Single Sourcing RCP/RAP (cont'd)



- consistent bundle hierarchy in application
- uncluttered code
- target environment is only a export question

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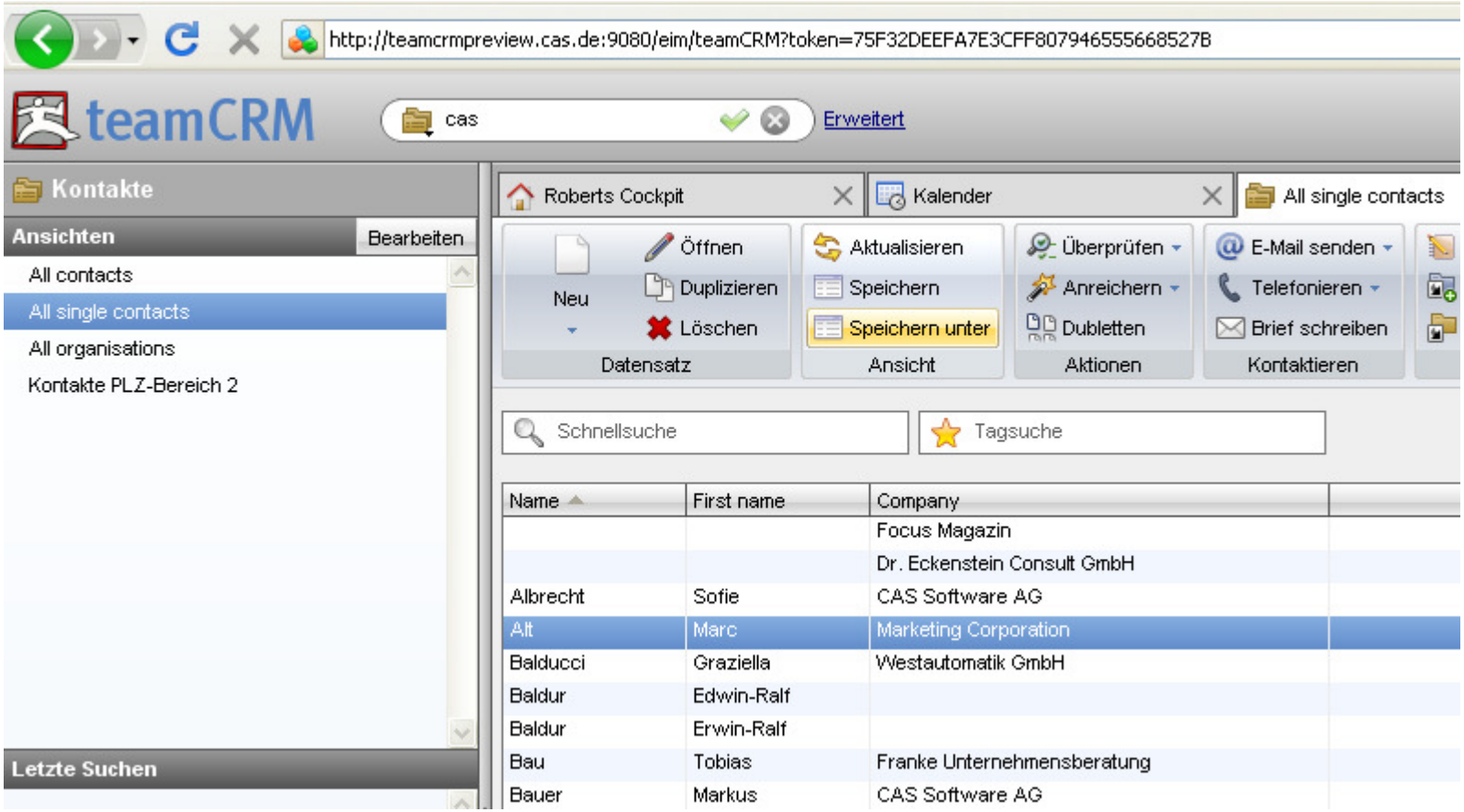
Agenda - Demo



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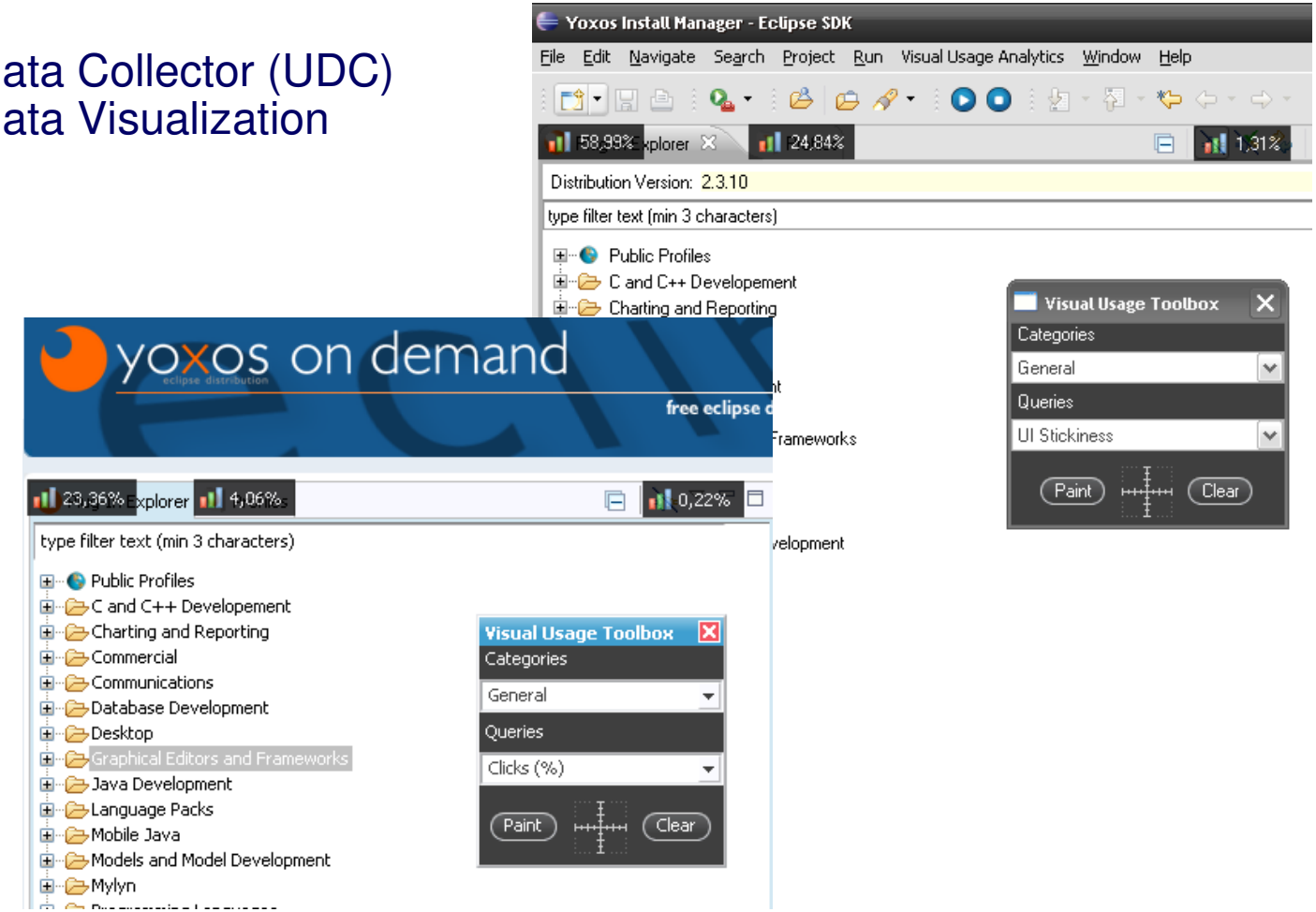
Usage Analysis

- Usage Data Collector (UDC)
- Usage Data Visualization



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Moving to the new Eclipse RT project

- the new Eclipse RT (Runtime, not Realtime) top level project helps to coordinate the runtime efforts
- the Equinox Community will foster and promote runtime technologies at Eclipse – independently of their location in a top level project
- RAP is one of the initial six projects of RT
- integration with other runtime technologies is key

http://eclipse.org/equinox-portal/

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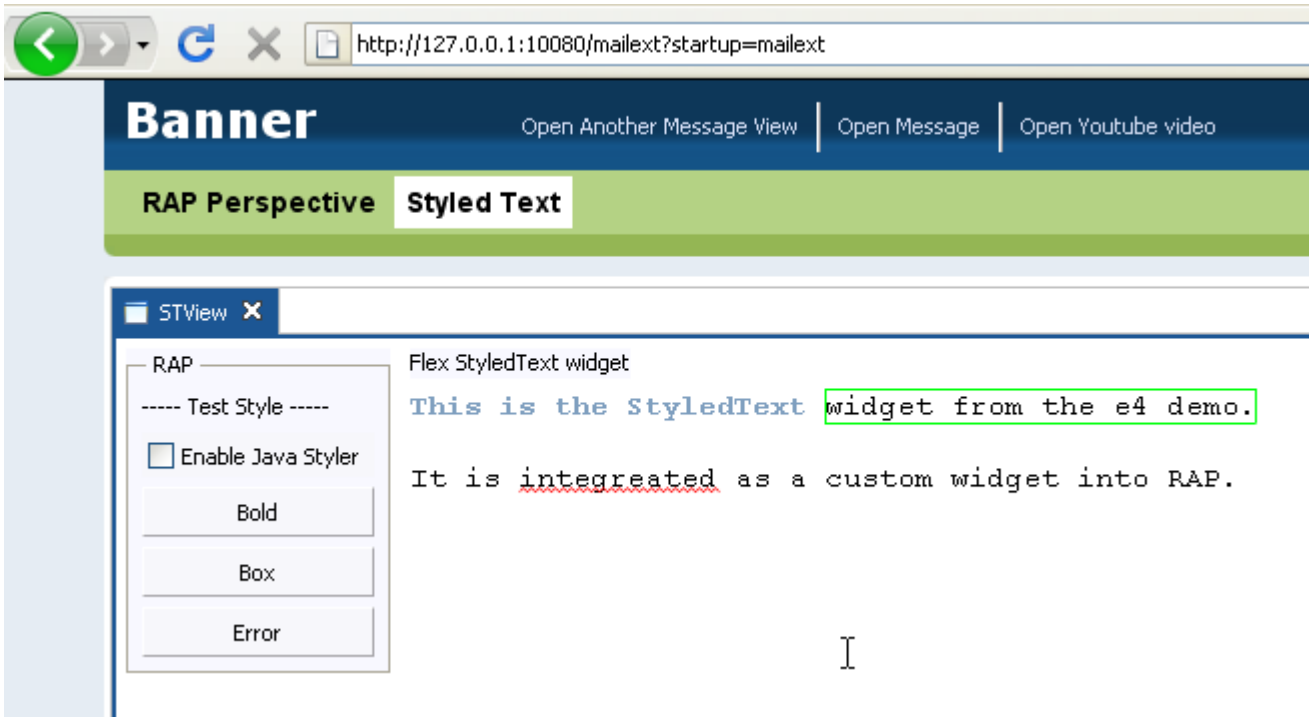
RAP & e4

- Platform and RAP team will work together on next generation of Eclipse platform, community is invited to join



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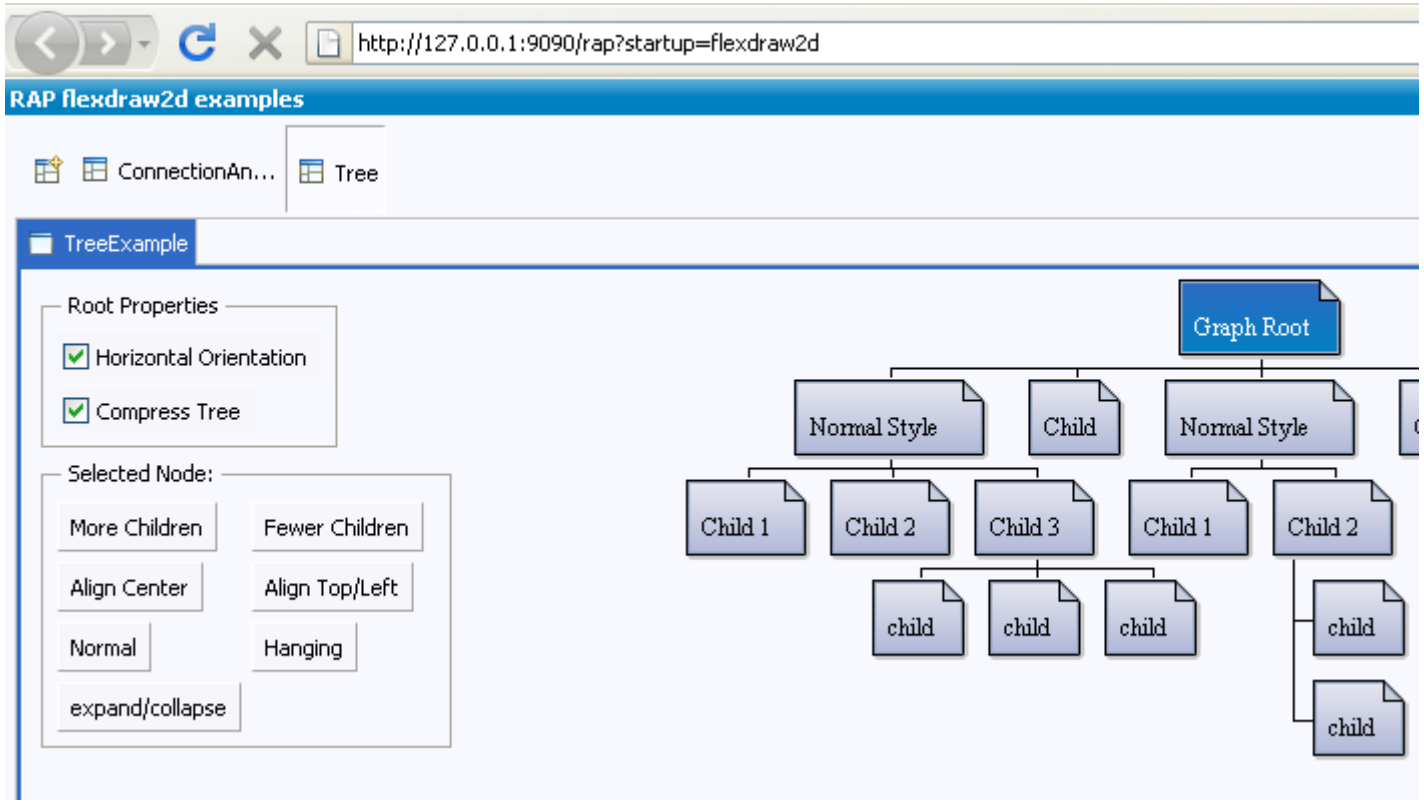
draw2d

- prototype using Flash in the browser



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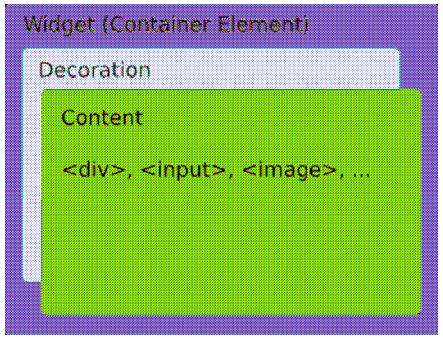
qooxdoo 0.8

- Decoration
  - customization mechanism of the widget's look & feel
  - support of rounded corners, shadow, etc.
  - widgets can have different decoration renderers



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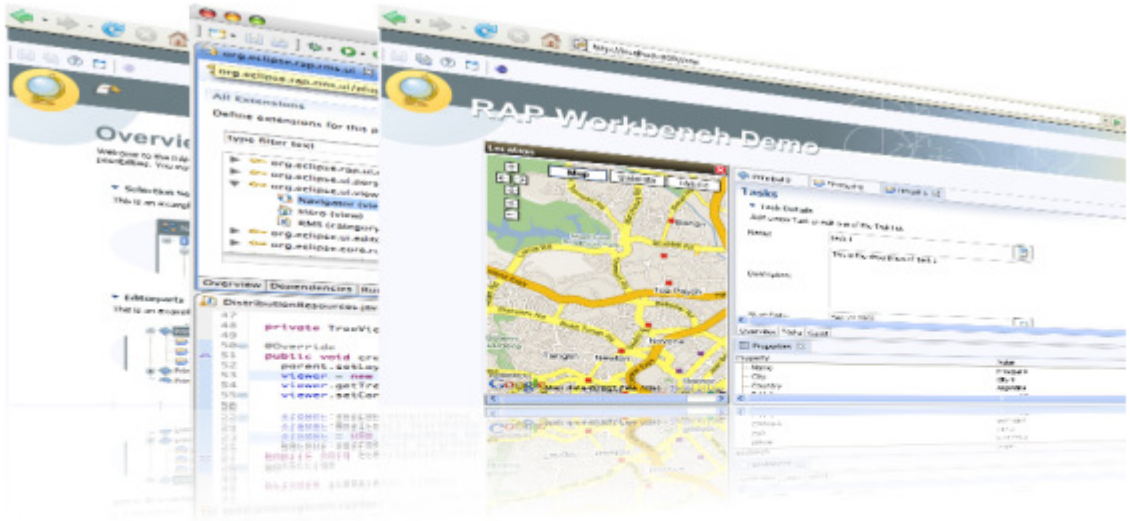
Get the RAP - http://eclipse.org/rap



Demos

See some demos here

Downloads

Get the latest RAP release

The RAP project enables developers to build rich, Ajax-enabled Web applications by using the Eclipse development model, plug-ins with the well known Eclipse workbench extenstion points, JFace, and a widget toolkit with SWT API (using **qooxdoo** for the client-side presentation). The project has graduated from incubation and released its 1.0 release.

Learn more ...

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References

- http://www.eclipse.org/rap - RAP project page
- http://wiki.eclipse.org/RAP - RAP project wiki
- http://www.qooxdoo.org - qooxdoo js library

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